QUALITY ASSURANCE VERIFICATION SUITE

Certificate Multi-Batch Isolation & 7-Day TTL Recovery

Target Branch: bagfix-mongoDB | Author: CodeX Core Engineering | Version: 2.1.0

Audience: Junior Developers, QA Engineers, and System Testers

PRIMARY TEST OBJECTIVES:

1. Verify supplementary batch edit/rejection does NOT delete main batch certificates (Regression Test).
2. Verify 7-day TTL soft-delete flow in MongoDB (revokes verification without permanent loss).
3. Verify Super Admin 1-click restore & hard purge from the 7-Day Trash Vault.

1. Test Execution Matrix

ID	TEST SCENARIO	PRIORITY	RESULT
TC-CERT-01	Multi-Batch Creation (Same Event & Date)	Critical	PASS [ ]
TC-CERT-02	Supplementary Batch Edit Isolation (Bug Fix)	Critical	PASS [ ]
TC-CERT-03	Batch Rejection Soft-Deletes Strictly Owned Records	Critical	PASS [ ]
TC-CERT-04	Public Verification Revoked on Soft-Delete	High	PASS [ ]
TC-CERT-05	Super Admin 1-Click Restore from 7-Day Trash	High	PASS [ ]
TC-CERT-06	Super Admin Immediate Hard-Delete & Empty Trash	Medium	PASS [ ]
TC-CERT-07	Duplicate Student Email Deduplication in CSV Import	High	PASS [ ]
TC-CERT-08	RBAC Security: Standard Admin Blocked from Trash	High	PASS [ ]

2. Step-by-Step Test Specifications for Junior / QA

TC-CERT-01 **Multi-Batch Creation (Same Event Name & Date)**

GOAL: Confirm multiple batches with identical event name and date can coexist.

1. Log in as Super Admin (superadmin@codex.community).
2. Go to Bulk Certificates > Direct Certificate Issuance.
3. Enter Event Name: "CodeX Orientation 2026", Event Date: "2026-09-03", Coordinator: "Parul Tyagi".

EXPECTED: Batch 1 is COMPLETED. Alice, Bob, and Charlie have valid certificates and active verification URLs.

TC-CERT-02 **Supplementary Batch Edit Isolation (Regression Bug Fix)**

GOAL: Verify editing a second batch with the same event name/date does NOT delete Batch 1 certificates.

1. As Admin, create Batch 2 with SAME Event: "CodeX Orientation 2026" and SAME Date: "2026-09-03".
2. Add 2 late students: David Miller (david@test.com), Emma Watson (emma@test.com). Submit for approval.
3. Click Edit (pencil icon) on Batch 2. Change coordinator name and add a 6th student.

EXPECTED: Batch 1 certificates remain 100% active and untouched (count >= 3). Batch 2 updates independently.

TC-CERT-03 **Batch Rejection Soft-Deletes Strictly Owned Certificates**

GOAL: Confirm rejecting a batch moves only that batch certificates to trash.

1. As Super Admin, click Approve on Batch 2. Wait until status becomes COMPLETED.
2. Now click Reject on Batch 2 with reason: "Late roster cancelled by coordinator".
3. Confirm rejection dialog in UI.

EXPECTED: Only Batch 2 certificates are moved to 7-day trash. Batch 1 certificates remain COMPLETED and active.

TC-CERT-04 **Public Verification Revocation on Soft-Delete**

GOAL: Confirm soft-deleted certificates cannot be verified on the public verification portal.

1. Copy the certificateId of soft-deleted student David Miller from Batch 2.
2. Open an incognito browser window: /verify-certificate/<DAVID_CERT_ID>.
3. In a separate tab, open Alice Green verification URL from Batch 1: /verify-certificate/<ALICE_CERT_ID>.

EXPECTED: David certificate displays "Invalid Certificate / Revoked". Alice certificate displays verified green badge.

TC-CERT-05 **Super Admin 1-Click Restore from 7-Day Trash Vault**

GOAL: Confirm Super Admin can instantly restore a certificate back to active status.

1. Log in as Super Admin. Navigate to Bulk Certificates > "7-Day Trash & Recovery" tab.
2. Observe counter badge and countdown timer (e.g. 6d 23h remaining auto-purged by TTL).
3. Locate David Miller row and click the green "Restore" button.

EXPECTED: David Miller vanishes from trash and his public verification URL immediately displays verified credential.

TC-CERT-06 **Super Admin Immediate Hard-Delete & Empty Trash**

GOAL: Verify Super Admin can bypass the 7-day TTL and permanently delete records.

1. In "7-Day Trash & Recovery", click the red trash can icon on Emma Watson row to hard-delete.
2. Click "Empty Trash" at top right to purge all remaining soft-deleted items.
3. Check MongoDB: db.certificates.countDocuments({ isDeleted: true }).

EXPECTED: Items are permanently deleted immediately (countDocuments === 0). UI displays "Trash is Empty".

TC-CERT-07 **Duplicate Student Email Deduplication in CSV Import**

GOAL: Verify CSV uploads containing duplicate student emails are deduplicated before submission.

1. Prepare CSV with 5 rows where 2 emails are repeated (3 unique students total).
2. In Bulk Certificates, click Import CSV and select the file.

EXPECTED: Toast indicates: "Successfully imported 3 unique students (2 duplicate emails deduplicated)". Exactly 3 certificates generated.

TC-CERT-08 **RBAC Security: Standard Admin Blocked from Trash Vault**

GOAL: Confirm standard Admins cannot view or restore certificates in the 7-day recovery vault.

1. Log in as standard Admin (admin@codex.community). Check tab bar in Bulk Certificates.
2. Try navigating to /admin/certificates?tab=trash or calling GET /api/v1/certificates/trash.

EXPECTED: Trash tab is completely hidden. Direct URL shows "Access Restricted". API call returns 403 Forbidden.

3. QA SIGN-OFF & DEPLOYMENT AUTHORIZATION

Tester Name: _____ Signature: _____ Date: _____

Build / Commit: bagfix-mongoDB Environment: [ ] Staging [ ] Pre-Prod [ ] Production